

broker-worker

Read-only access to your MetaTrader account, so the daily loss limit is enforced against real equity instead of a figure you typed in weeks ago.

It cannot place an order. No route in this Worker reaches a broker's trade endpoint, in either mode.

Two modes. The free one is the default.

	PUSH — FREE	META-API — PAID
Cost	£0	MetaApi subscription
How	A Python script beside your own terminal pushes snapshots	A hosted terminal is polled
Current when	Your PC and MT5 are running	Always
Needs	Cloudflare free tier	Cloudflare + MetaApi

Everything below is the free path. If you later want it current with your machine off, set `MODE` to `metaapi` and follow the short section at the end.

What you need

- A Cloudflare account (free)
- MetaTrader 5 on a PC, logged into your broker
- Python

Nothing else, and no card.

Setup

1. Storage and secrets

```
cd tools/broker-worker
npx wrangler login
npx wrangler kv namespace create STORE
```

Paste the id it prints into `kv_namespaces` in `wrangler.jsonc`, and put your account number in `ALLOWED_ACCOUNTS`.

Then invent a long random string — this is the shared key between your PC and your Worker, so make it unguessable:

```
npx wrangler secret put PUSH_KEY
npx wrangler deploy
```

ALLOWED_ACCOUNTS is not optional and the Worker refuses to serve without it. The account id arrives from outside — from the browser on a read, from the pusher on a write — so an empty list means anyone who finds the URL can write whatever they like into your daily-limit check.

2. The desk-side script

On the PC running MetaTrader:

```
pip install MetaTrader5 requests
```

```
set BE_PUSH_URL=https://bullionedge-broker.<subdomain>.workers.dev
set BE_PUSH_KEY=<the same value you gave wrangler>
set BE_ACCOUNT=<your MT5 account number>
python mt5_push.py
```

It prints a line every two minutes with your equity and trade count. Leave it running while you trade — a scheduled task or a shortcut in **shell:startup** saves remembering.

3. Point the app at it

Settings → **Broker sync** → on. Proxy endpoint is the Worker URL, account id is your MT5 login. Press **Test connection**.

Check it yourself

```
curl "https://<your-worker>.workers.dev/account?account=<id>"
```

Then check it refuses what it should:

```
curl -X POST "https://.../push?account=<id>" -d '{}'
```

 # 401, no key

```
curl "https://.../account?account=not-yours"
```

 # 404, not listed

```
curl -X POST "https://.../account?account=<id>"
```

 # 405, read route

```
curl "https://.../trade?account=<id>"
```

 # 404, no such route

The refused-account answer is identical to the missing-route answer, so the endpoint cannot be used to work out which account ids are real.

What the push key can and cannot do

If it leaks, someone can write **wrong numbers into your own daily-limit check**. That is the whole blast radius: the write path stores a snapshot and never contacts a broker, so no order can be placed and no money can move. Rotate it with `npx wrangler secret put PUSH_KEY` and restart the script.

Your MetaTrader password is never involved. The script attaches to the terminal session you have already logged into; the only thing that leaves your machine is balance, equity and closed trades.

The honest limitation

Snapshots are only as fresh as the last run of the script, so this is accurate while your PC is on and stale afterwards. The app knows: it refuses any snapshot older than five minutes rather than showing a number that has stopped being true. Stale equity is worse than none — it reads as authoritative and will happily report room to trade after the loss that used the room up.

The part that is easy to get wrong

MetaTrader reports **deals, not trades**. One position produces an entry deal and an exit deal, and only the exit carries the realised profit. Importing them one-for-one doubles every trade — an entry booked as a break-even and an exit booked as the whole result — which halves the win rate while leaving both numbers looking entirely plausible.

The exit deal also carries the **opposite type** to the position: you close a buy by selling. Reading direction off it files every winning long as a winning short.

`pairDeals()` groups by position, takes direction from the entry, close price and time from the last exit, and sums profit, commission and swap across every leg so partial closes and both-sided commission are counted. Both modes go through it, so there is one implementation rather than two that can disagree.

Covered by `test/worker-pairing.test.mjs` and `test/worker-push.test.mjs`, including a round-trip test asserting that what this Worker emits is what `domain/broker.js` accepts.

Order ticket

Separate feature, no proxy needed at all. The desk and the planner both produce the exact order — symbol, type, volume, entry, stop, targets — for you to type into your terminal. Set your broker's symbol for gold in Settings; brokers spell it `XAUUSD`, `GOLD`, `XAUUSD.m`, `XAUUSDpro`, and a ticket carrying the wrong one has to be retyped.

If you want the paid mode later

Set `MODE` to `metaapi`, add your account in MetaApi (investor password, never the master one), `npx wrangler secret put METAAPI_TOKEN`, set

`METAAPI_REGION` to the region MetaApi provisions you in, and redeploy. The KV namespace and the push route go unused; the app needs no change.

Costs

Cloudflare Workers free tier: 100,000 requests a day. KV free tier: 100,000 reads and **1,000 writes** a day — which is why the pusher defaults to 120 seconds (720 writes) and refuses to run under 60 (1,440, and your afternoon stops working).